

Ramzy Bakir

ramzybakir.rsb@gmail.com | 01278673139 | Heliopolis, Cairo, Egypt

www.linkedin.com/in/ramzy-bakir | <https://ramzybakir.github.io> | <https://github.com/RamzyBakir> | <https://huggingface.co/RamzyBakir>

WORK EXPERIENCE

AI Engineer – Mercury Visual Solutions; Cairo, Egypt December 2025 – July 2026

- **Built** and **improved** a **multi-agent AI workflow system** for managing creative/project planning tasks.
- Developed **agents** using **LangGraph** orchestration, **ReAct** patterns, **tool** calling, **memory**, and **prompt-driven workflows**.
- Built **backend AI services** with **FastAPI**, **Pydantic**, **SQLAlchemy**, **Alembic**, and **async APIs**.
- Developed **backend AI agent infrastructure** with **persistent** chat sessions, **memory** handling and **tool** based execution.
- **Designed** and implemented **full-stack chatbots** and **workflows** including **prompt** optimization, **query** generation **improvements**, **model configuration**, **guardrails**, and response **safety rules**.
- Enhanced **AI workspaces UI** features with improved **layouts**, **responsive components**, entity **management** actions, navigation updates, and visual polish.

Data Scientist – Seif Pharmacies Group; Cairo, Egypt April 2025 – May 2026

- Built **automated ETL** pipelines to ingest, clean, normalize, and structure raw attendance and HR **data**.
- Developed **statistical analysis pipelines** to compute employee performance metrics with **report generation pipelines**.
- Engineered **clustering algorithms** to segment employees into performance buckets based on **KPIs**, feedback scores, and **behavioral data**.

Converged Data Solution Architect Intern- Orange Egypt; Smart Village, Egypt August 2024 - September 2024

- Collaborated with senior **solution architects**, understanding the structure of the **data centers** running **virtual machines**, maintaining, optimizing and deploying **docker containers** and **Kubernetes**.
- Processed and wrangled **1 million records** representing **30k** users' internet usage sessions.
- Developed and presented a dashboard featuring **24 KPIs** and **24 graphs**, providing actionable insights for bucket and offer recommendations.

RPA Business Automation Developer- Advansys ESC; Cairo, Egypt April 2024 - October 2024

- Contributed to **process documentation and automation project proposals**, improving clarity and adoption of RPA solutions for business clients
- **Developed invoice processing agent**, improving efficiency of financial workflows by automating extraction and validation from PDF invoices.

Data Analysis & Business Intelligence Intern - Centro; Cairo, Egypt September 2023 - October 2023

- Gained practical expertise in leveraging **statistical analysis** and **data visualization** tools, including **Excel**, **PowerBI**, and **Incorta**, to drive data-driven insights and optimize decision-making processes.
- Selected as one of the **top 3 interns** out of 12 for exceptional performance, particularly recognized for delivering the **best Power BI project**.

RPA Business Analyst Intern - Advansys ESC; Cairo, Egypt July 2023 - August 2023

- Developed expertise in **RPA business analysis**, with a focus on writing **PDDs**, creating **documentation**, and gathering client requirements.
- Created Process Definition Documents (**PDDs**) for approved automation projects.
- Initiated **two automation processes**, effectively managing **client meetings**, gathering requirements, conducting **business analysis**.

EDUCATION

EUI - Egypt University of Informatics November 2022 - June 2026

B.S. Computing and Information Sciences New Administrative Capital, Egypt

- **Fresh Graduate;** C-GPA: 3.13
- **Graduation Project:** TAQDIR - AI-Powered RFP Evaluation Platform | Developed for & under the mentorship of Advansys ESC.
 - **Grade:** A+
 - Served as **Team Lead & AI/ML Architect**, driving the system architectural iterations, design decisions and managing 5 team members.
 - Maintained **integration** and **alignment** between the **software engineering** and **AI teams** throughout the project **lifecycle**.
 - **Led** the **AI/ML architecture** and **development** of the **agentic RFP evaluation workflows**, **owning** the complete **AI evaluation** pipeline from initial **architecture** through **four major architectural** iterations and **production-ready**.
 - Built the **end-to-end document intelligence pipeline**, chunk-based extraction, document reshaping/normalization, **Arabic text processing**, and **OCR services integration**.
 - **Architected** a **LangGraph-based conversational AI chat agent** with recruitment/procurement-specific **tools**, **conversation memory**, **state management**, **streaming**, and **safety controls**.
 - Implemented **end-to-end observability** using **OpenTelemetry**, **Tempo**, and **Grafana**, including distributed **trace propagation** across **AI evaluations** and **Celery workflows**, automated **smoke tests**, datasource provisioning, and comprehensive **technical documentation**.
 - Built **core backend services** for **orchestration**, **processing**, **scoring analytics**, **Redis/Celery-based** progress reporting.
- **Relevant courses:** Introduction to Machine Learning, Agentic AI, MLOPS, Natural Language Processing, Computer Vision, Introduction to AI, Big Data Engineering, Algorithms Design and Analysis, Advanced Programming, Introduction to Software Engineering, and Software Design and Development, Computational Linear Algebra, Introduction to Probability and Statistics, and Calculus I & II.

Collège de la Salle / Lycée Simone de Beauvoir Cairo, Egypt

French Baccalaureate September 2008 – July 2022

PROJECTS & PUBLICATIONS

TlntCo - <https://tlntco.com/>

February 2026- Present

LangGraph, Celery, Next.js, Docker

- Served as **Full Stack AI Engineer** in the development of a **production SaaS** recruitment platform, from **architecture** to **deployment** and active usage.
- Project garnered national recognition with features on **national TV** and radio alongside coverage by **32 national** and **regional news outlets** along side being **IP-registered**.
- Built and maintained a **LangGraph-based AI Chat agent** - Talo with **tool** usage, **memory**, **context**, **streaming** responses, **conversation** tracking, and recruitment-specific **guardrails**.
- Designed AI-driven resume screening and candidate **evaluation workflows** with bulk generation, document parsing, **Pydantic** structured output.
- Engineered async task orchestration using **Celery** with **RabbitMQ** and **Redis** with retry policies, rate limiting, and TTL-based retention.

Unitree Go2 — AI Voice-Controlled Quadruped Robot

May 2026 – June 2026

Python, FastAPI, ROS2, Whisper, Unitree SDK, DDS

- Built a real-time **AI voice-control** system for the **Unitree Go2 robotics platform**, integrating **Whisper speech recognition**, **natural-language command** parsing, **WebSockets**, and **DDS-based robot control**.
- Mapped **30+ robot behaviors** with **live state streaming** and **mock hardware testing**, gaining hands-on experience with robotics middleware, real-time control, and robot communication.

Arabic Embeddings & RAG

June 2025 – May 2026

RAG, LLMs, Sentence Embeddings, Python

- Fine-tuned **Alibaba-NLP/gte-multilingual-base** using **LoRA** on the **Silma Arabic Triplets Dataset (~2million samples)**.
- Supports a wide range of downstream NLP tasks: **semantic search**, **semantic similarity**, **paraphrase mining**, **text classification**.
- One of the top ranking models **on the Arabic RAG Leaderboard**, highlighting its effectiveness for **Retrieval-Augmented Generation pipelines** in Arabic with **~300 monthly** downloads
- View and use the models on **HuggingFace** via: <https://huggingface.co/RamzyBakir/arabic-gte-multilingual-embed-medium>

AI & ML Security & Phishing Detection Suite

May 2025- December 2025

PyTorch, HuggingFace Transformers, LoRA

- Built an **end-to-end phishing detection ecosystem**, from curating a **2.5M-sample URL dataset** to **training and deploying transformer models** for URL, email, and LLM-based cybersecurity threat analysis. Model List:
 - o Model1: [HuggingFace Link](#) | Model2: [HuggingFace Link](#) | Model3: [HuggingFace Link](#)

Ancient Egyptian NLP Model Family

April 2025 - May 2025

Generative & Classificatory Modeling of Hieroglyphic Transcriptions

Python, HuggingFace Transformers

- **AegyptusTranslit1-gpt2-17M & roberta-aegyptustranslit-classifier**
 - o Designed and trained a **17M parameter GPT-2 architecture-based model** from **scratch** for generating Ancient Egyptian transliterations.
 - o Using a dataset of **~ 30,000** sentences of transliterated Ancient Egyptian; built a **BPE tokenizer** with a vocabulary size of **6,475 tokens**. Implemented full **training pipeline** including data preparation, model training, evaluation, and text sampling.
 - o View and use the model on **HuggingFace** via: <https://huggingface.co/RamzyBakir/AegyptusTranslit1-gpt2-17M>
 - o View and use the model on **HuggingFace** via: <https://huggingface.co/RamzyBakir/roberta-aegyptustranslit-classifier>

SKILLS AND CERTIFICATES

- **Programming Languages:** Python, R, C, C++, SQL, Scala
- **Data Visualization:** Matplotlib, Seaborn, Power BI, Tableau, Incorta, ggplot2, Tidyverse
- **Data Cleaning & Transformation:** Pandas, NumPy, Beautiful Soup
- **ML Algorithms:** Time Series Analysis, Regression Analysis, Gradient Boosting, KNN, Decision Trees, CNNs, Neural Network, Transformers, Supervised Fine-Tuning (SFT), LoRA, PEFT, Evaluation & Benchmarking
- **AI / ML Frameworks & Tools:** Scikit-learn, XGBoost, TensorFlow, PyTorch, LightGBM, CatBoost, Optuna, Hugging Face, LangChain, LangGraph, LangSmith, LangFuse, Pydantic, Streamlit, Document Processing.
- **LLMs & NLP:** Experience with developing, building and finetuning LLMs, LLM serving, agentic workflows, MCP and RAG.
- **Statistical Analysis:** SciPy, A/B Testing, Hypothesis Testing, Bayesian Methods
- **Observability:** OpenTelemetry, Grafana, Tempo, AI Traceability & Monitoring.
- **SWE, Ops & Automation:** UiPath, Microsoft Power Automate, Git, Jira, Docker, Jenkins, Nginx, CI/CD Pipelines.
- **Languages:** Arabic (Native), English & French (Fluent), German (Conversational)
- **Certificates:** [Duke University Open Source LLMops Solutions](#) | [UiPath Academy Automation Business Analyst](#) | [IBM Data Science Specialization](#) | [Data Scientist with Python - Datacamp](#) | [Machine Learning with Python - IBM](#) | [Data Analysis with R - IBM](#)

❖ You can access all the aforementioned project details, including source code and documentation, within my GitHub repositories via the following link: <https://github.com/RamzyBakir?tab=repositories>